

Name

Adrianna Milton

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Gender

Non-binary

Residence: City (State), Country

South Royalton (Vermont), United States of America

Nationality

American

When did you or do you expect to complete your PhD?

2024

Please tell us briefly about your university education (institution, city, country, year)

I hold two university degrees earned in the United States. I completed my Bachelor of Science in Psychology at Florida State University in Tallahassee, Florida, USA in 2014, where I gained foundational training in behavioral and physiological psychology, including hands-on neuroanatomy instruction and undergraduate research in taste perception and memory.

I then pursued doctoral training in Neuroscience at Case Western Reserve University in Cleveland, Ohio, USA, graduating in 2023. My PhD research focused on spinal cord injury and neural repair, during which I developed surgical models of chronic cervical spinal cord injury and investigated combinatorial treatments aimed at restoring forelimb function. This work was conducted in the Department of Neuroscience at the School of Medicine, and contributed to peer-reviewed publications and national conference presentations.

Why would you like to be a Science Corps Fellow?

Science has always been, for me, both a discipline and a practice — a way of observing the world with curiosity, rigor, and openness. Becoming a Science Corps Fellow represents an opportunity to extend that practice beyond the laboratory and into communities that stand to benefit most from accessible, engaging, and human-centered science communication. It is a chance to do something I have long believed in: make science feel like it belongs to everyone.

Throughout my career as a neuroscientist, I have been driven not only by discovery but by the conviction that science loses much of its power when it remains confined to academic spaces. The questions I have pursued — how the injured nervous system can be coaxed toward recovery, how the brain processes reward and smell, how zebrafish can teach us about the gut-brain axis — are questions that matter to people far beyond the walls of a research institution. Yet the language, culture, and infrastructure of academic science can make that relevance invisible to the very public whose curiosity and support sustains it. I want to change that.

I have worked throughout my career to bridge that gap. Through guest lectures at Dartmouth and STEM outreach with S.P.A.R.K. at local Vermont schools, I have seen firsthand what happens when science is presented not as a body of facts to absorb, but as a living, human process of asking and exploring. Those experiences taught me that effective science communication is less about simplification and more about invitation — drawing people into the questions, not just the answers.

What distinguishes my approach is the integration of mindfulness focusing on mental health/wellbeing into how I communicate and teach. In my role as a Mindfulness Instructor at Dartmouth's Student Wellness Center, I have come to understand that people engage more deeply with difficult or unfamiliar ideas when they feel present, grounded, and respected. Attention is not a given — it is something that must be cultivated and honored. I bring this philosophy into every scientific space I inhabit, from college lecture halls to public school cafeterias, and I would carry it into my work as a Science Corps Fellow.

I am equally motivated by what Science Corps represents for scientists like me. As a Black woman in neuroscience, I have spent my career navigating spaces where my presence was not assumed or expected. That experience has made me a more empathetic communicator, a more purposeful mentor, and a sharper advocate for the idea that science communication must actively reflect the full diversity of those who do science — and those who deserve access to it. Representation in who tells scientific stories matters as much as representation in who conducts the research. A fellowship with Science Corps would allow me to act on that conviction with intention, structure, and reach as well as facilitate a mutually beneficial partnership.

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

Some of my most meaningful work as a scientist has happened outside the laboratory — in classrooms, hallways, and community centers, sitting alongside students who were encountering science, perhaps for the first time, as something that could belong to them.

Over the course of my graduate and postdoctoral career, I have mentored more than a dozen students, ranging from high schoolers navigating their first research experience to undergraduates developing independent projects in neuroscience. Through formal programs at Case Western Reserve University — including the Scientific Enrichment Opportunity Program, the Northern Ohio AGEP Alliance, and the Provost Scholars Program — I provided one-on-one after-school mentorship to students from underrepresented and economically at-risk backgrounds in Cleveland, Ohio. Many of these mentees were young women of color for whom seeing a scientist who looked like them was itself a meaningful part of the experience. I never took that lightly.

More recently, as a postdoctoral researcher at Dartmouth, I have continued that work through the Summer Undergraduate Research Fellowship and the Presidential Scholars Program, mentoring students from institutions across New England. These relationships reminded me that good mentorship is not just about teaching techniques or explaining concepts — it is about helping someone believe that they are capable of asking hard questions and finding answers. Alongside mentoring, I have been deeply involved in community STEM outreach through S.P.A.R.K. (STEM Programs for Aspiring Researchers and Kids) at Dartmouth, where I served as Vice President for Education. Through S.P.A.R.K., I helped bring science programming to students at Ottauquechee School in White River Junction, Vermont, and to a Girls STEM Club at Middlebury Middle School — spaces where curiosity was abundant but access to scientific role models was not. These visits were a reminder of how much young people hunger for science when it is presented with enthusiasm, warmth, and relevance to their lives.

Together, these experiences have shaped how I think about science engagement: not as outreach performed at a distance, but as relationship-building grounded in genuine care for the people in the room. That is the spirit I would bring to every interaction as a Science Corps Fellow.

Where do you imagine yourself professionally in the future? Check all that apply.

In science education
In science outreach

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.

My professional future lies at the intersection of research, education, and outreach — and I see discipline-based education research (DBER) as the thread that ties all three together. I do not imagine leaving science; I imagine expanding what doing science means, and deepening how it is taught and experienced.

In the near term (~2 years), my immediate goal is to develop a research program that applies the rigor of empirical inquiry to questions about how people learn science — particularly students from underrepresented backgrounds. As I complete my postdoctoral fellowship at Dartmouth, I am building a foundation in DBER while also exploring the role of contemplative pedagogy in science education. Contemplative practices — including mindfulness, reflective writing, and intentional presence — have long been used in the humanities and social sciences to deepen learning, but their application in STEM education remains underexplored. I believe this is a significant and timely gap. My dual background in neuroscience and mindfulness instruction uniquely positions me to investigate how contemplative approaches can reduce anxiety, increase belonging, and improve engagement in science classrooms — especially for students who have historically felt excluded from scientific culture. A Science Corps Fellowship would strengthen this trajectory by grounding my education and outreach work in real communities and real relationships, informing the questions I bring to my research.

In the long term (~10 years), I see myself as a faculty member whose work sits at the intersection of neuroscience, DBER, and contemplative pedagogy — leading a research program that asks not only how the brain works, but how we can create the conditions in which all students can learn about it most fully and freely. I want to build a lab that produces both neuroscientific discoveries and evidence-based educational frameworks that integrate contemplative practice into STEM teaching in meaningful, measurable ways. Alongside that research, I envision an outreach program embedded in my institution and the broader community — one that models the same warmth, presence, and inclusivity that contemplative pedagogy brings to the classroom.

Ultimately, I want to be the kind of scientist and educator whose students — whether they are doctoral researchers, undergraduates, or middle schoolers at a community science night — leave feeling more capable, more grounded, and more certain that science is a place where they belong.

How did you find out about Science Corps?

An email list in your institution
